Chebyshev's Inequality Summary

L22.pdf
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Lecture 22 Scribe – [Exact Topic Name from Slide 1]

2. Key Definitions & Notation
List all definitions exactly as introduced
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Preserve wording and mathematical accuracy
3. Core Concepts & Logical Development
Present concepts in the same order as the lecture
Clearly show how each concept builds on previous ones
Include:
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Conditions of validity
Important observations stated in the lecture
4. Step-by-Step Derivations / Proofs

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5. Worked Examples
Reproduce all examples from the lecture
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6. Logical Dependencies

Explicitly indicate relationships between concepts
Show how definitions → theorems → results → applications connect
7. Exam-Focused Summary
Concise recap of:
Key formulas
Definitions
Results most likely to be tested
No new explanations — only what exists in the material

Reasoning Guidelines:

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Produce a single, well-organized lecture scribe document such that:

A student can fully revise Lecture 22 using only this document.

Lecture 22 Scribe – Chebyshev's Inequality & Tail Bounds

(Based strictly on Lecture 22 slides L22)

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1. Key Definitions & Notation

Random Variable & Moments

- X: random variable
- $\mu = E[X]$: mean
- $\mathrm{Var}(X) = E[(X-\mu)^2]$: variance
- $\sigma = \sqrt{\mathrm{Var}(X)}$: standard deviation

Markov's Inequality

For a non-negative random variable X and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's Inequality

For any random variable X with mean μ and variance $\mathrm{Var}(X)$:

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Tail Event

- Event of the form:

$$|X - \mu| \geq a$$

- Represents values **far from the mean**

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2. Core Concepts & Logical Development

Motivation

- Markov's inequality provides a bound using only the mean
- Chebyshev improves this by incorporating **variance**
- Variance controls **spread around the mean**

Key Idea

- Tail probability depends on how much spread exists
- Larger variance → larger probability of deviation
- Smaller variance → tighter concentration around mean

Logical Flow

1. Start with Markov inequality
2. Apply it to a transformed variable
3. Use squared deviation from mean
4. Obtain Chebyshev inequality

Concept Progression

- Markov: bound using X

- Chebyshev: bound using $(X-\mu)^2$
- Later extension (introduced): exponential transformation e^{tX}

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3. Step-by-Step Derivation of Chebyshev's Inequality

Step 1: Start with probability of deviation

$$P(|X-\mu| \geq a)$$

Step 2: Rewrite event

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Step 3: Apply Markov's Inequality

Let $Y = (X-\mu)^2$, which is non-negative:

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

Step 4: Substitute expectation

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

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4. Worked Examples

Example 1: Coin Toss (Running Example)

Setup:

- $X \sim \mathrm{Binomial}(n, \frac{1}{2})$
- $\mu = \frac{n}{2}$

Tail Event:

$$P\left(X \geq \frac{3n}{4}\right)$$

- Represents values far from mean (right tail)

Key Insight:

- Chebyshev gives a **better bound than Markov**

Example 2: Exam Scores (Real-Life Example)

Given:

- Mean $\mu = 70$
- Standard deviation $\sigma = 10$
- Variance $\sigma^2 = 100$

Question:
Fraction of students below 40 or above 100:

$$|X - 70| \geq 30$$

Apply Chebyshev:

$$P(|X - 70| \geq 30) \leq \frac{100}{30^2}$$
$$= \frac{100}{900} = \frac{1}{9}$$

Result:

- At most $\frac{1}{9}$ of students are far from mean

Example 3: Exam Grades with Variance

Given:

- Mean $\mu = 70$
- Standard deviation $\sigma = 5$
- Variance $\sigma^2 = 25$

Step 1: Convert condition

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

Step 2: Apply Chebyshev

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

$$P(|X - 70| \geq 20) \leq \frac{25}{400}$$

Result:

$$P(X \geq 90) \leq \frac{1}{16}$$

Example 4: Network Latency SLA

Given:

- $\mu = 50$ ms
- $\mathrm{Var}(X) = 16$

Failure condition:

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Apply Chebyshev:

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$
$$= \frac{1}{25}$$

Result:

- At most **4%** packets fall outside range

Example 5: Manufacturing Tolerances (Homework)

Given:

- Mean $\mu = 100\Omega$
- Defective if outside $[94, 106]$
- $a = 6$

Part (a):

$$P(|X - \mu| \geq 6) \leq \frac{\sigma^2}{6^2}$$

Part (b):

- Multiply probability bound by 10,000 to get expected count

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5. Logical Dependencies

Dependency Chain

1. **Definition of mean and variance**
   ↓
2. **Markov's inequality (on non-negative variables)**
   ↓
3. **Transformation:** $(X-\mu)^2$
   ↓
4. **Chebyshev's inequality**
   ↓
5. **Applications to tail probabilities**

Extension Introduced

- Markov: uses X
- Chebyshev: uses $(X-\mu)^2$
- Chernoff (introduced idea): uses e^{tX}

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6. Key Idea: Exponential Tail Bounds

Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Insights

- Exponential function grows very fast
- Large values of X become much larger
- Tail events are amplified
- Leads to tighter bounds

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7. Exam-Focused Summary

Key Formulas

- Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- Chebyshev:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Key Steps for Problems

1. Identify μ and σ^2
2. Convert condition to:

$$|X-\mu| \geq a$$

3. Apply Chebyshev formula
4. Substitute values
5. Simplify fraction

Important Observations

- Chebyshev works for **any distribution**
- Uses only **mean and variance**
- Provides **upper bound**, not exact probability
- Tighter than Markov when variance is known

Common Patterns

- Convert inequalities into absolute deviation
- Square appears due to variance
- Always results in **fraction form bound**

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Lecture 22 Scribe – Chebyshev's Inequality

(Based strictly on Lecture 22 slides L22)

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1. Key Definitions & Notation

Random Variable and Moments

- X: random variable
- $\mu = E[X]$: mean
- $\mathrm{Var}(X) = E[(X-\mu)^2]$: variance
- σ: standard deviation, where $\sigma^2 = \mathrm{Var}(X)$

Markov's Inequality

For a non-negative random variable X and $a > 0$:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev's Inequality

For any random variable X with mean μ:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Equivalent Form Used in Derivation

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Tail Event

- Event of the form:

$$|X - \mu| \geq a$$

Exponential Transformation (for tail bounds)

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

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2. Core Concepts & Logical Development

Idea: Using Mean and Variance

- Mean $+$ variance controls deviation from mean
- Variance measures spread

From Markov to Chebyshev

- Markov inequality bounds probability using X
- Chebyshev inequality bounds probability using $(X - \mu)^2$

Conceptual Progression (as stated)

- Markov: bound using X
- Chebyshev: bound using $(X - \mu)^2$
- Chernoff: bound using e^{tX}

Key Observations

- Tail corresponds to values far from mean
- Squaring transforms deviation into a non-negative quantity
- Applying Markov to squared deviation yields tighter bound

Conditions of Validity

- Markov requires non-negative random variable
- $(X - \mu)^2 \geq 0$, so Markov is applicable

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3. Step-by-Step Derivation of Chebyshev's Inequality

Step 1: Start with deviation probability

$$P(|X-\mu| \geq a)$$

Step 2: Rewrite using square

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Step 3: Apply Markov's Inequality

Let:

$$Y = (X-\mu)^2$$

Then:

$$P(Y \geq a^2) \leq \frac{E[Y]}{a^2}$$

Step 4: Substitute expectation

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

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4. Worked Examples

Example 1: Running Example (Coin Flips)

Setup:

- $X \sim \mathrm{Binomial}(n, \frac{1}{2})$
- $\mu = \frac{n}{2}$

Tail Event:

$$P\left(X \geq \frac{3n}{4}\right)$$

- Tail represents values far from mean

Example 2: Exam Scores (Real-Life Example)

Given:

- Mean $\mu = 70$
- Standard deviation $\sigma = 10$
- Variance $\sigma^2 = 100$

Step 1: Convert condition

$$X \leq 40 \text{ or } X \geq 100$$

$$\Rightarrow |X - 70| \geq 30$$

Step 2: Apply Chebyshev

$$P(|X - 70| \geq 30) \leq \frac{100}{30^2}$$

Step 3: Simplify

$$= \frac{100}{900} = \frac{1}{9}$$

Result:

- At most $\frac{1}{9}$ students are far from mean

Example 3: Solved Example 1 (Exam Grades with Variance)

Given:

- Mean $\mu = 70$
- Standard deviation $\sigma = 5$
- Variance $\sigma^2 = 25$

Step 1: Identify parameters

$$\mu = 70, \quad \sigma = 5 \Rightarrow \sigma^2 = 25$$

Step 2: Convert condition

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

Step 3: Apply Chebyshev with $a = 20$

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Step 4: Substitute values

$$P(|X-70| \geq 20) \leq \frac{25}{400}$$

Result:

$$P(X \geq 90) \leq \frac{1}{16}$$

Example 4: Solved Example 2 (Network Latency SLA)

Given:

- $E[X] = 50$ ms
- $\mathrm{Var}(X) = 16$

Step 1: Identify parameters

$$\mu = 50, \quad \sigma^2 = 16$$

Step 2: Rewrite failure condition

$$X \notin [30, 70] \Rightarrow |X-50| \geq 20$$

Step 3: Apply Chebyshev

$$P(|X-50| \geq 20) \leq \frac{16}{400}$$

Step 4: Simplify

$$= \frac{1}{25}$$

Result:

- At most 4% packets fall outside range

Example 5: Homework Problem (Manufacturing Tolerances)

Given:

- Mean $\mu = 100\Omega$
- Defective if outside $[94, 106]$

Step (a): Express condition

$$|X - \mu| \geq 6$$

Apply Chebyshev:

$$P(|X - \mu| \geq 6) \leq \frac{\sigma^2}{6^2}$$

Step (b): Expected count

- Multiply probability bound by 10,000

---

5. Logical Dependencies

Dependency Structure

1. Definition of mean and variance
2. Markov's inequality (requires non-negativity)
3. Transformation: $(X - \mu)^2$
4. Application of Markov → Chebyshev inequality
5. Application to tail events

Progression of Bounds

- Markov → Chebyshev → Chernoff

---

6. Exponential Tail Bound Concept

Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Stated Insight

- Exponential grows very fast
- Large values of X become much larger
- Tail events get amplified
- Leads to tighter tail bound

---

7. Exam-Focused Summary

Formulas

- Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Standard Procedure

1. Identify μ, σ^2
2. Convert condition to:

$$|X - \mu| \geq a$$

3. Apply Chebyshev inequality
4. Substitute values
5. Simplify

Important Results from Lecture

- $P(|X - 70| \geq 30) \leq \frac{1}{9}$
- $P(X \geq 90) \leq \frac{1}{16}$
- $P(|X - 50| \geq 20) \leq \frac{1}{25}$

Key Points

- Uses mean and variance
- Works for any random variable
- Produces upper bound on tail probability
- Requires transformation into absolute deviation form

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- $\mathrm{Var}(X) = E[(X-\mu)^2]$: variance
- σ: standard deviation, where $\sigma^2 = \mathrm{Var}(X)$

Markov Inequality

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Chebyshev Inequality

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Equivalent Form Used

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Exponential Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

---

3. Core Concepts & Logical Development

Idea

- Uses mean + variance
- Variance controls deviation from mean

Progression of Ideas

- Markov: bound using X
- Chebyshev: bound using $(X-\mu)^2$
- Chernoff: bound using e^{tX}

Important Observations

- Tail = values far from mean
- Squared deviation ensures non-negativity
- Enables application of Markov inequality

Conditions of Validity

- Markov applies to non-negative random variables
- $(X-\mu)^2 \geq 0$, hence applicable

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4. Step-by-Step Derivation of Chebyshev Inequality

Step 1

$$P(|X-\mu| \geq a)$$

Step 2

$$P(|X-\mu| \geq a) = P((X-\mu)^2 \geq a^2)$$

Step 3

Apply Markov inequality:

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

Step 4

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

---

5. Worked Examples

Example 1: Running Example (Coin Flips)

Setup

$$X \sim \mathrm{Binomial}(n, 1/2), \quad \mu = \frac{n}{2}$$

Tail

$$P\left(X \geq \frac{3n}{4}\right)$$

Example 2: Exam Scores

Given

$$\mu = 70, \quad \sigma = 10 \Rightarrow \sigma^2 = 100$$

Step 1

$$X \leq 40 \text{ or } X \geq 100 \Rightarrow |X-70| \geq 30$$

Step 2

$$P(|X-70| \geq 30) \leq \frac{100}{30^2}$$

Step 3

$$= \frac{100}{900} = \frac{1}{9}$$

Example 3: Solved Example 1

Given

$$\mu = 70, \quad \sigma = 5 \Rightarrow \sigma^2 = 25$$

Step 1

$$X \geq 90 \Rightarrow |X-70| \geq 20$$

Step 2

$$P(|X-70| \geq 20) \leq \frac{25}{400}$$

Result

$$P(X \geq 90) \leq \frac{1}{16}$$

Example 4: Solved Example 2 (Network Latency)

Given

$$\mu = 50, \quad \sigma^2 = 16$$

Step 1

$$X \notin [30, 70] \Rightarrow |X-50| \geq 20$$

Step 2

$$P(|X-50| \geq 20) \leq \frac{16}{400}$$

Step 3

$$= \frac{1}{25}$$

Result

- At most 4% packets outside range

Example 5: Homework Problem

Given

$$\mu = 100, \quad |X - \mu| \geq 6$$

(a)

$$P(|X - \mu| \geq 6) \leq \frac{\sigma^2}{36}$$

(b)

- Multiply by 10,000

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6. Logical Dependencies

Flow

Mean, Variance
→ Markov inequality
→ Transform to $(X - \mu)^2$
→ Apply Markov
→ Chebyshev inequality
→ Apply to tail bounds

Hierarchy of Bounds

- Markov → Chebyshev → Chernoff

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7. Exam-Focused Summary

Formulas

$$P(X \geq a) \leq \frac{E[X]}{a}$$

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Standard Steps

1. Identify μ, σ^2
2. Convert to $|X - \mu| \geq a$
3. Apply Chebyshev
4. Substitute
5. Simplify

Results

$$\frac{1}{9}, \quad \frac{1}{16}, \quad \frac{1}{25}$$

Key Points

- Uses mean and variance
- Gives upper bound
- Works for any random variable

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